Does agreement among sector ETF peers predict price catch-up?

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1. Hypothesis Pitch

When a sector or industry ETF fails to move while its historically related sector or industry ETF peers move together in one direction, it tends to catch up over the next five trading sessions. I select its five peers from the other ETFs using the highest correlations of daily returns over the previous 252 trading observations, ending before the signal day. Specifically, I think the catch-up is stronger when the peer move reflects agreement across several peers than when the same average move comes from one exceptional peer. I measure agreement as the standard deviation of the five peers returns on the signal day, divided by the peers average daily volatility over the previous 21 sessions. The lower it is, the more alike the peers moved, relative to how much they usually move. I assume that T's five-day catch-up should be larger when that standard deviation is close to 0, and smaller when it is high.

Example

Both columns have the same +2% average peer return, and the standard deviation is 0% in scenario 1 and about 4.5% in scenario 2. I predict the target T rises, and it rises more where the standard deviation is 0%.

Metric	Scenario 1: Broad Agreement	Scenario 2: One Exceptional Peer
Peer A's return today	+2%	+10%
Peer B's return today	+2%	0%
Peer C's return today	+2%	0%
Peer D's return today	+2%	0%
Peer E's return today	+2%	0%
Average of the 5 peers	+2%	+2%
Spread of the 5 peers (standard deviation)	0%	4.5%
Target ETF T's return today	0.2%	0.2%
T's return over the next 5 days (my prediction)	Large catch-up: e.g. +1.4%	Small catch-up: e.g. +0.3%

Scaled by the peers normal daily volatility (say 1%), the spreads become 0 and 4.5, so Scenario 1 would be Low dispersion and Scenario 2 High.

Why I think it might be true

When several related sector or industry ETFs rise by similar amounts, this may reflect shared economic information. My proposed mechanism is that investors adjust exposure to some sectors before others, temporarily leaving target ETF T behind. However, different sector sensitivities could also explain the divergence, so a lag is not automatically evidence of delayed reaction.

When the same average peer return comes mainly from one exceptional ETF, the rise may instead reflect information specific to that sector. There is then less reason for T to follow. This is why I expect stronger catch-up when the spread of peer returns is lower. I will measure each trade's return relative to its five peers and compare the average for Low and High dispersion signals, checking that both groups had similar peer moves. The test uses upward peer moves only.

What would prove me wrong

I would abandon the hypothesis if across the out of sample period, trades after low dispersion peer moves do not catch up and do not even more over five sessions than trades after high-dispersion peer moves.
I would also abandon its use as a trading strategy if the catch up happens before I can enter at the next trading session’s open, or if transaction costs remove the benefit. If the results are too uncertain to draw a clear conclusion, I would report the evidence as inconclusive.

The target and its peers are drawn from a fixed sample of 21 sector and industry SPDR ETFs, all launched before the test begins. This tests sector-level catch-up, rather than individual-company catch-up. The funds share market exposure, so signals across funds and overlapping five-session periods must not be treated as independent evidence.

2. Data Pipeline

Data Retrieval

I used Yahoo Finance, via the yfinance Python library, to obtain daily price data for a fixed sample of 21 sector and industry ETFs. The sample includes nine broad sector ETFs and twelve industry ETFs covering areas such as banking, biotechnology and semiconductors. This provides a manageable universe of related assets for constructing peer groups. All 21 ETFs launched by June 2006, according to their State Street fact sheets. The requested download period runs from 1 December 2009 through 31 December 2025. The intended backtest begins in January 2012, because the earlier observations provide the history needed before the first trade. 2010 is used to estimate the correlations that choose the first peer groups (252 trading days), and 2011 to calibrate the Low/High dispersion cutoffs for 2012. One extra month (December 2009) ensures a full 252 daily returns exist before 2011, because the first price has no return. Each ETF is downloaded separately. The download uses auto_adjust=True, which adjusts Open, High, Low and Close prices for splits and distributions. These adjusted prices are subsequently used to calculate returns. Adjusted prices are revised retrospectively whenever a distribution occurs.

The retrieval process checks that each download contains data, includes a closing-price column and has at least one nonmissing closing price. Failed downloads are reported, and the pipeline stops if any ETF fails, preventing download problems from silently changing the universe. Successful downloads are combined into one table, aligned by date, with each column identifying both the price field and ETF.
Before cleaning, the downloaded data are saved in CSV and Parquet formats to preserve the input snapshot. By default, the notebook reloads this Parquet snapshot instead of downloading again, so it runs without internet access and reproduces the reported results exactly. Because Yahoo revises adjusted prices after each distribution, a fresh download (USE_SNAPSHOT = False) can produce slightly different values.

Data Cleaning

I clean a copy of the downloaded data so that the original snapshot remains unchanged. Volume is removed because the strategy uses only Open, High, Low and Close prices. The pipeline checks that every ETF has these fields, converts dates and prices to the appropriate types if needed, and sorts observations chronologically. Exact duplicate dates or columns are removed, while conflicting duplicates stop the pipeline for investigation. Failed numeric conversions are recorded.
Zero, negative and infinite prices are replaced with NaN. The pipeline also checks whether High is below Low or whether Open or Close falls outside the day's low–high range. A tolerance of 0.01% is allowed for rounding when comparing Open and Close with those boundaries. If an OHLC contradiction is detected, all four prices for that ETF-date are marked as NaN because the code cannot identify which value is incorrect. The affected observations and reasons are recorded.
Missing prices are checked within each ETF's observed history, with the first and last available dates recorded separately for each field. Missing values are not filled because our strategy is return based, and ETFs without usable closing prices are excluded with an explanation. These checks use dates already present in the dataset rather than an independently verified trading calendar.
Additional checks flag movements above +20% or below −20% and runs of at least five identical closing prices. These are review flags rather than automatic deletion rules.
The cleaning output reported no issues from the other checks (0 duplicates, 0 conversion errors, 0 invalid prices, 0 OHLC conflicts, 0 missing values, 0 stale runs) .
Seven large movements were flagged across four ETF-date observations:

ETF	Date	Movement	Return
XES	9 March 2020	Previous Close → Close	−33.72%
XES	9 March 2020	Previous Close → Open	−25.06%
XLE	9 March 2020	Previous Close → Close	−20.14%
XOP	9 March 2020	Previous Close → Close	−36.87%
XOP	9 March 2020	Previous Close → Open	−24.24%
XOP	10 March 2020	Previous Close → Close	+21.84%
XOP	10 March 2020	Previous Close → Open	+28.22%

All three affected ETFs have energy-sector exposure: XLE covers energy companies, XOP focuses on oil and gas exploration and production, and XES covers oil and gas equipment and services.
The dates coincide with the March 2020 oil-price shock, and prices were verified to be true on TradingView. The observations were retained rather than removed solely for exceeding the threshold. The cleaned OHLC dataset is saved separately in CSV and Parquet formats.

3. Strategy and Backtest

Signal Generation

The signal-generation stage identifies ETFs that remain approximately flat while their historically correlated peers rise on average. It then separates these observations according to how closely the peers move together. The parameter choices described below are justified in Methodology Defense.
Before calculating signals, the code checks that the cleaned closing-price columns contain exactly the predefined ETF universe. If cleaning excluded an ETF, execution stops so that the sample does not change silently. Prices and returns are then stored in NumPy arrays, with rows representing dates and columns representing ETFs. Daily returns are calculated from adjusted closing prices using pct_change(fill_method=None), which leaves missing observations unfilled.
The find_peers() function selects five peers for each ETF at the beginning of each month. It uses the previous 252 trading observations, ending before the first session of that month, to calculate a Pearson correlation matrix. For each target ETF, the function excludes the target itself, ranks the other ETFs by their signed correlation, from highest to lowest, and selects the five highest. These peers remain fixed throughout the month. The function stops if the historical window contains fewer than 252 observations or any missing returns.
The find_candidates() function examines each trading session using that month's selected peers. It calculates the equally weighted average return of the five peers and keeps observations where this average is at least +1%. This version therefore studies catch-up following upward peer moves. Each candidate is an ETF-day observation, and one target ETF on one date. A single trading session can contribute several candidates if the peers of several target ETFs meet the threshold.
At this stage, the target does not need to be a laggard. The broader set of candidate observations is also used to calculate the historical dispersion thresholds.
Peer agreement is measured using the sample standard deviation of the five peers' returns on the signal day. This is divided by the peers' average recent volatility:

Scaled dispersion = standard deviation of the five peer returns ÷ average 21-session trailing volatility of the five peers.

Each peer's trailing volatility is the standard deviation of its daily returns over the previous 21 sessions. The .shift(1) operation excludes the signal day from this calculation. For example, a peer-return standard deviation of 0.5%, divided by average trailing volatility of 2%, produces a scaled dispersion of 0.25. A lower value indicates that the peers' returns are more similar relative to their usual daily fluctuations.
The label_signals() function then identifies laggards. A target qualifies when its signal-day return is between −0.25% and +0.25%, inclusive, while its peers' average return meets the +1% threshold. This gives a concrete definition of an ETF that has remained approximately flat despite a positive average move among its peers.
For each test year, the Low and High dispersion thresholds come exclusively from the previous calendar year's candidate observations. These observations are pooled across ETFs and include all qualifying peer-move observations, regardless of whether the target was a laggard. The lower threshold is the 33⅓ percentile of scaled dispersion, and the upper threshold is the 66⅔ percentile. Current-year laggards are labelled **Low** when their dispersion is at or below the lower threshold and High when it is at or above the upper threshold. Observations between the thresholds are excluded.
For example, the 2012 thresholds are calculated using candidates from 2011. They remain unchanged throughout 2012, while peer selection is updated monthly. Because the thresholds come from an earlier year and the laggard condition applies an additional filter, the current year's signals do not necessarily divide into equal thirds.
The output contains the signal date, target and peer identifiers, average peer return, target return, scaled dispersion and Low/High classification. The code also displays signal counts by year and group, together with a calibration table showing the number of previous-year candidates and the two thresholds used. These outputs make the signal selection and annual calibration visible.
Signals depend on the signal day's closing prices, so they become available only after that session closes. Entry timing, the five-session holding period, position sizing and portfolio performance are handled in the subsequent backtest. Generated signals therefore do not necessarily correspond to executed trades.
Over 2011–2025, the peers' average return reached at least +1% on 13,868 ETF-day observations. From these candidates, 351 laggard signals fell into the Low or High groups between 2012 and 2025: 139 Low and 212 High. Across years, the calibrated Low thresholds ranged from 0.25 to 0.66, and the High thresholds ranged from 0.42 to 0.97. Some years were particularly unbalanced, for example 2012 produced 2 Low and 17 High signals, 2018 produced 25 Low and 3 High, and 2021 produced 9 Low and 34 High. These differences limit year-by-year comparisons, particularly for the 2012 development year, which contains only two Low signals.

Backtesting

The backtest uses two independent long only portfolios: one trades Low-dispersion signals and the other High-dispersion signals. Each starts with $100,000 and follows the same rules, running continuously from 2012 to 2025 with cash and positions carried across months and years.
Each trade is a Position object storing the target ETF, its peers, its dispersion group and the original signal information. A position enters at the open of the session after the

signal and exits at the close of the fifth session, counting the entry session as the first (entry_time + HOLD - 1). The Portfolio class manages cash, open positions, closed trades and daily equity. Open positions are stored in a dictionary keyed by ETF, so checking whether an ETF is already held is a single lookup. Only one position per ETF is allowed.
Each new position receives 20% of current equity, valued at that session's opening prices, with fractional units permitted. A trade proceeds only if available cash covers the purchase and its entry cost, which prevents borrowing. In practice, cash never constrained a trade: of 351 signals, 342 became trades (135 Low, 207 High), and the nine skipped signals were all for ETFs already held. Transaction costs are charged on both entry and exit, at 5, 10 and 20 basis points per side in separate runs, with 5 basis points used for the headline results. Uninvested cash earns no interest.
run_backtest() processes sessions in chronological order. In the morning, it enters signals generated at the previous session's close. In the evening, it closes positions whose holding period has ended, then values the portfolio at closing prices and records equity and the fraction invested. A trade is admitted only if its full holding period fits before the end of the data. Because every signal is acted on at the next open, and position sizes use morning prices, no trade uses information unavailable at the time.
Besides net profit, each position records its peer-relative return which is the target's return from entry open to exit close, minus the equally weighted return of its five peers over the same interval. A positive value means the target outperformed its peers. This is the catch-up measure, calculated before costs, and because the peers' return is subtracted, general market movements cancel out. Each position also records the same measure from the signal close to the entry open (gap_relative_return). This separates catch-up that happened overnight, before the strategy could enter, from catch-up during the holding period.

period_stats() reports results separately for 2012 (in-sample) and 2013–2025 (out-of-sample):
cumulative return, CAGR, Sharpe ratio (zero risk-free rate, 252 sessions per year), maximum drawdown, average exposure and trade statistics.
Portfolio returns follow their calendar dates, while each trade is assigned to the year of its signal.
The hypothesis test uses out-of-sample trades from the 5-basis-point portfolios and subtracts the High group's average peer-relative return from the Low group's, weighting each trade equally. A positive difference would support the hypothesis. Because trades overlap in time and the ETFs move together, they are not independent observations. bootstrap_low_minus_high() therefore resamples whole calendar months of trades, 1,000 times with a fixed seed, and reports the 2.5th–97.5th percentile range as a 95% interval. Dependence between trades in different months is not fully captured.

Results:

Table 1. Portfolio performance (5 basis points per side)

Period	Group	Cumulative return	CAGR	Sharpe	Max drawdown	Avg. exposure	Trades
In-sample 2012	Low	−0.8%	−0.8%	−1.25	−0.8%	0.6%	2
In-sample 2012	High	+1.6%	+1.6%	0.70	−1.4%	5.1%	16
Out-of-sample 2013–2025	Low	+21.4%	+1.5%	0.49	−5.3%	3.3%	133
Out-of-sample 2013–2025	High	+12.8%	+0.9%	0.37	−8.3%	4.7%	191

Table 2. Trade-level results, out-of-sample

Group	Win rate	Mean net trade return	Mean peer-relative return	Mean overnight relative return	Signal-day peer move	Scaled dispersion
Low	60.9%	+0.75%	−0.03%	−0.02%	1.34%	0.30
High	58.6%	+0.33%	+0.11%	−0.02%	1.41%	1.00

Hypothesis test: Low minus High peer-relative return = **−0.14%**, 95% bootstrap interval **[−0.61%, +0.27%]**.

Table 3. Transaction-cost sensitivity, out-of-sample

Cost per side	Low: return / Sharpe / max DD	High: return / Sharpe / max DD
5 bps	+21.4% / 0.49 / −5.3%	+12.8% / 0.37 / −8.3%
10 bps	+18.3% / 0.42 / −5.3%	+8.6% / 0.26 / −9.3%

20 bps	+12.1% / 0.29 / −5.4%	+0.6% / 0.03 / −11.4%

The hypothesis is not supported. It predicted that targets would catch up more after Low-dispersion peer moves than after High-dispersion ones. Out of sample, the difference has the opposite sign (−0.14%), and the 95% interval includes zero. The data provide no evidence that peer agreement predicts stronger catch-up, and they are inconsistent with any effect larger than about +0.3 percentage points.

There is little catch-up at all, as the average peer-relative return is close to zero in both groups: −0.03% for Low and +0.11% for High. Targets that stayed flat while their peers rose did not go on to outperform those peers over the following five sessions. The overnight relative return is also about zero (−0.02%), so no catch-up was missed before the strategy could enter. The failure is not a matter of entry timing.

We can say that the portfolio profits come from market drift, not the signal. Both portfolios gained out of sample, and Low had the higher return and Sharpe ratio. However, mean net trade returns (+0.75% and +0.33%) are far larger than the peer-relative returns, so most of the profit reflects ETF prices rising generally while positions were held. A long-only strategy would capture this whether or not the signal worked. The Low portfolio's advantage therefore does not come from stronger catch-up.

The groups are comparable, and the in-sample year is uninformative. Low and High signals followed similar peer moves (1.34% and 1.41%) and differed as intended in dispersion (0.30 and 1.00), so the comparison is not driven by move size. The 2012 in-sample year contains only two Low trades, so it cannot support a meaningful in-sample versus out-of-sample comparison.

Returns are small and sensitive to costs. Signals are infrequent, so average exposure is only 3–5% of capital, and annual returns are 0.9–1.5%. At 20 basis points per side, the High portfolio's out-of-sample return falls to almost zero (+0.6%, Sharpe 0.03), while Low remains positive (+12.1%, Sharpe 0.29).

4. Methodology Defense

The Universe is 21 SPDR sector and industry ETFs. Sector and industry ETFs are liquid, have long histories and contain naturally related assets, which makes peer groups meaningful. Testing at the sector level also reduces the influence of single-company news. All 21 funds launched by June 2006, so each has a complete history over the sample. If wrong, several funds overlap heavily (KBE and KRE; XOP, XES and XLE), so some peer agreement is mechanical rather than informative. A fixed list chosen today also carries survivorship risk as shown in 5.

Peers are top five by correlation over 252 days, recomputed monthly**.** One year of daily returns gives enough observations for stable correlations while staying recent. Peers are ranked by signed correlation, because a strongly negative relationship is not agreement in direction. Peer sets change slowly, so monthly updates lose little information and need 180 correlation matrices instead of about 3,770. If wrong*,* after a sudden regime change, peers can be out of date for up to a month.

Signal thresholds: peers' average return of at least +1%, target within ±0.25%**.** A 1% average move is roughly a full typical day's move for a diversified ETF, so it marks a genuine shared move. A return within ±0.25% is close enough to zero to describe a target that "failed to move". Both thresholds were set before testing. If wrong, the thresholds use raw returns and ignore each ETF's sensitivity to its peers. Defensive sectors such as utilities and consumer staples naturally move less, so they qualify as flat more easily: XLU, XLP and XLV make up 25% of High signals but only 16% of Low signals.

Volatility-scaled dispersion. The raw standard deviation of peer returns mostly measured market stress: in 2020 it placed 38 signals in High and only 4 in Low. Dividing by the peers' average volatility over the previous 21 sessions measures disagreement relative to how much the peers normally move. With scaling, 2020 split evenly (18 Low, 17 High). I chose this measure over two alternatives (dividing by the size of the peer move, and a 63-session window) by checking which was least related to market volatility and move size. I did not look at trade outcomes when choosing.

Previous-year terciles. Low and High cutoffs are the one-third and two-thirds points of the previous year's dispersion values, frozen before each year begins. This uses no future data, avoids hand-picking a threshold, and adapts if typical dispersion drifts. Excluding the middle third keeps the two groups clearly separated. If wrong, an unusual previous year unbalances the groups, as in 2018 (25 Low, 3 High).

Long-only. An earlier long/short version mixed market drift into the comparison: short trades fight the market's upward drift while long trades benefit from it, so Low and High differed partly by their mix of longs and shorts. Using upward peer moves only gives both groups the same drift exposure. The cost is downward catch-up is not tested.

Peer-relative return as the test. Catch-up is measured as the target's return minus its peers' average return over the same holding window. Movements shared by the target and its peers cancel, so the measure isolates whether the target closed the gap. If wrong, it is not adjusted for beta, so a low-sensitivity target can lag its peers in a rising market without any failure of catch-up.

In-sample 2012, out-of-sample 2013–2025. 2012 was examined while designing the strategy, so it is reported separately. The thirteen out-of-sample years include several distinct market regimes (2015–16, 2018, 2020, 2022). One limitation is 2012 contains only two Low trades, and design changes were made after early results (Section 5).

Costs: 5 basis points per side, stress-tested at 10 and 20. These ETFs are highly liquid, with narrow bid–ask spreads, and trades execute at the open. If wrong, the smaller industry ETFs (e.g. XES, KCE, XSD) trade with wider spreads. At 20 basis points the High portfolio's return falls almost to zero.

Sizing: 20% of equity per trade, no leverage. There are never more than three signals on the same day, and each trade lasts five sessions, so this size almost never runs out of cash. In practice cash never prevented a trade. Signals are rare, so average exposure is only 3–5% and absolute returns are small. The Sharpe ratio is unaffected by this scaling.

Month-block bootstrap. Trades overlap in time and the ETFs move together, so treating trades as independent would understate uncertainty. Resampling whole calendar months keeps nearby trades together. A fixed seed makes the interval reproducible. Limitation: dependence between trades in adjacent months is not fully captured.

5. Pitfall Checklist

Look-ahead bias
I went through every input to a signal and looked at whether I would know it at the close of the signal day. The peers are chosen from the 252 trading days before the month starts, so the month itself is never included. The volatility used to scale dispersion is shifted back by a day, so the signal day's own move doesn't affect it. The Low/High cutoffs only use the previous year's data. On the trading side, a signal is only known after the close, so the earliest the backtest can act on it is the next morning's open, and position sizes are based on that morning's prices.
I don’t have any point where the strategy uses information it wouldn't have had.

Survivorship bias
My universe is a fixed list of 21 ETFs that all still exist today, so any sector or industry ETF that closed or merged during 2009–2025 is not in it. I didn't try to rebuild which funds were available in each year. Newer sector funds like XLRE (2015) and XLC (2018) are also left out, because they don't have enough history for the peer calculation. The bias is smaller here than it would be with individual stocks because an ETF's price already reflects every company it held along the way, including ones that were later delisted, acquired or went bankrupt. ETFs that close tend to do so because nobody is buying them, not because their price collapsed. If I had included them, I'd expect noisier peer groups and higher trading costs, because those funds were less liquid.

Overfitting/data-snooping
I didn't optimise any parameter for returns. The only thing the code calibrates, the dispersion cutoffs, is re-estimated each year from the year before. However, I did change the design several times before settling on this version: I expanded the universe from 9 to 21 ETFs, moved from long/short to long-only, replaced raw dispersion with volatility-scaled dispersion (after testing two other scalings), simplified the comparison and switched to monthly peers, changed position size from 1/21 to 1/5 of equity, and tried a +2% peer-move threshold once before returning to +1%. Some of these changes came after I had seen results covering 2013–2025, so that period isn't a perfectly clean holdout. Each change removed a specific problem rather than improving performance, and the final result is still negative, so data-snooping is unlikely to explain it.

Transaction costs and slippage
Every trade pays costs on the way in and out. I used 5 basis points per side for the main results, because these are liquid ETFs with tight spreads and the trades happen at the open. I also re-ran everything at 10 and 20 basis points. I found costs make a

difference, and out of sample, the Low portfolio's return drops from +21.4% to +12.1% at 20 basis points, and High's falls from +12.8% to +0.6%.
The bigger issue is that the strategy isn't attractive even before costs. The peer-relative return, which is the part that should come from catch-up, is close to zero, so whatever money it makes comes from the market rising while positions are open. The backtest also assumes I always get exactly the opening price with no market impact. That's optimistic for the smaller industry ETFs like XES, KCE and XSD, whose spreads are wider than 5 basis points.

Out-of-sample validation

I looked at 2012 while designing the strategy, so I report it as in-sample and treat 2013–2025 as out-of-sample, with the cutoffs re-calibrated each year from the previous one. In 2012 the Low portfolio lost 0.8% on just two trades, and High gained 1.6% on sixteen. Out of sample, Low returned +21.4% (Sharpe 0.49) on 133 trades and High +12.8% (Sharpe 0.37) on 191.
What matters more is that neither period shows what the hypothesis predicted. Low didn't catch up more than High in either: the difference was −0.30% in 2012 and −0.14% out of sample, with a 95% interval of −0.61% to +0.27%. I wouldn't put much weight on 2012, though, because two Low trades is far too few to compare. The out-of-sample years weren't completely untouched while I was making design decisions.

6. Efficiency Touchpoint

The slowest step in my pipeline is the bootstrap, which takes about 1.5 seconds. Each of the 1,000 resamples rebuilds a full sample of the 324 out-of-sample trades by stitching randomly chosen months back together, so the cost is

$$O(B \cdot n): B \times n = 1{,}000 \times 324 = 324{,}000\ row\ copies.$$
$$n = \ number\ of\ trades\ being\ resampled$$
$$B = \ Number\ of\ resamples$$

Most of the time goes on pandas overhead, because it builds a new table for every resample. At this scale it causes no problems and executes fast. If I needed many more resamples or trades, I would precompute each month's total and trade count for the Low and High groups. Each resample would then only add up month totals, which cuts the cost to

$$O(B \cdot M),$$
$$where\ M\ = \ the\ number\ of\ months$$

The step I deliberately made cheaper is choosing peers. A correlation matrix across N ETFs over L days costs O(L·N²). Recomputing peers every day would mean about 3,770 matrices. Peer groups barely change from one day to the next, so I recompute them once a month instead, which needs only 180 matrices and takes about 0.01

seconds in total. That saving would matter with a bigger universe, for example with 500 stocks instead of 21 ETFs, each matrix would be over 500 times more expensive.
I also used dictionaries where lookups happen inside the daily loop. Open positions are keyed by ETF, and signals are keyed by day, so checking if we already hold this certain ETF or if there were signals the day before is O(1), not a search through a list.

7. Risk Touchpoint

The strategy never borrows. Each trade gets 20% of equity, and a new trade only goes ahead if there is cash to pay for it. Because signals are rare, the portfolios hold cash most of the time: Low is invested on only 13% of trading days and High on 18%, and average exposure is just 3–5%. But that average hides how concentrated the risk is when it does appear. Signals cluster on strong market days, so positions tend to open together. At its peak, Low held three positions at once (61% of equity) and High held four (80%).
Those positions are also less diversified than they look. They are all long, they were all triggered by the same kind of broad rally, and many of the ETFs overlap in holdings. KBE, KRE and XLF are all banks and financials, and XOP, XES and XLE are all energy. Some of the most frequently traded funds are volatile industry ETFs like XME, XES and XBI, so three or four open positions can effectively be one bet on the market or on one sector.
A plausible losing scenario is a short-lived rally inside a falling market. Peers jump by more than 1% on a relief day, the strategy buys several laggards the next morning, and then a volatility spike pushes everything down within the five-day hold. With 80% of equity invested in correlated longs, a 10% market drop would cost roughly 8% of the portfolio in a week, larger than Low's worst drawdown (5.3%) and close to High's (8.3%). Single sector shocks already show the scale: the worst trades were XES (−15.4%, in 2022) and XHB (−11.0%, in 2024).

8. Reflection

This project reminded me of my final-year project, an indicator-based backtester. In that system, a strategy is written as rules, parsed into an abstract syntax tree (AST), and evaluated against price data to generate entry and exit signals. I reused some of its structure here, for example a Position class for each trade, a Portfolio that manages cash and open positions, and a loop that moves through the data one day at a time. I also used the pandas skills I have developed from this project.

The biggest lesson was how easily something else can pass for a signal. Raw dispersion looked like a measure of peer agreement, but it turned out to be mostly a measure of market stress. In 2020 it put 38 signals in High and only 4 in Low. Even in the final version, both portfolios made money out of sample, but the peer-relative return

was close to zero. The profit came from the market rising while positions were held, not from catch-up, and I would have missed that if I had judged the strategy by its returns alone. I also learned that overlapping trades are not independent evidence, so a simple average says little without a block bootstrap behind it.

I assumed the basic catch-up existed and that peer agreement would strengthen it. However, flat ETFs did not go on to outperform their peers, whatever the dispersion. I also assumed that flat while peers rise meant the target T was lagging. In practice, many of these signals came from defensive sectors like utilities and staples, which naturally move less than their peers. One possible explanation is that sector ETFs are heavily traded and arbitraged, so any delayed reaction is too small or too quick to show up over five days.

I originally wanted to test this on individual stocks. Using today's S&P 500 members would have introduced survivorship and look-ahead bias, because it leaves out companies that were removed or went bankrupt, and includes companies that only joined later. I found historical lists of index members, but couldn't download prices for many of them because tickers had changed or been delisted. So I moved to ETFs. That brings its own limits: a fixed list of funds that exist today, overlapping holdings, and no beta adjustment.
The sample is small, with 342 trades, so effects below about 0.3 percentage points can't be detected.
The in-sample year has only two Low trades, and the out-of-sample period was not untouched while I made design changes.
Only upward moves and one holding period were tested.

Realistic next steps could be:

1. Beta-adjusted definitions: Define "lagging" relative to the target's expected move given its sensitivity to its peers, and measure catch-up the same way.
2. Using stocks: Return to individual stocks using a source with permanent identifiers that survive ticker changes and delistings.
3. Isolating catch-up directly. Test downward moves, and a market-neutral version (buy the target, sell its peers) that removes market drift by construction, accounting for short-selling costs.

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